「Thm. Riemann's Theorem.

Let $f$ be a Complex function having an isolated singularity at $z=z_0$.
If $f$ is bounded for some deleted neighborhood of $z_0$,
then $z_0$ is removable.

Proof. By the hypothesis, for some $R>0$, $f$ is analytic and bounded in

$$0<|z-z_0|<R.$$

Given $z_1 \in B_R(z_0) \setminus \{0\}$, let $r>0$ s.t. $r<|z_1-z_0|<R$. So, $f$ is analytic and bounded in the annulus between $C: |z-z_0|=R$ and $C_1: |z-z_0|=r$.

By the Cauchy's integral formular,

$$f(z_1)=\frac{1}{2\pi i}\cdot\left[\int_C \frac{f(u)}{u-z_1}\,du - \int_{C_1}\frac{f(u)}{u-z_1}\,du\right]$$

Meanwhile, by the ML inequality,

$$\left|\int_{C_1}\frac{f(u)}{u-z_1}\,du\right| \le \sup|f|\cdot 2\pi r\cdot\frac{1}{|z_1-z_0|-r} \to 0 \quad \text{as } r\to 0.$$

So, $$f(z_1)=\frac{1}{2\pi i}\int_C\frac{f(u)}{u-z_1}\,du.$$

Since $z_1$ was arbitrary, we can denote that

$$f(z)=\frac{1}{2\pi i}\cdot\int_C\frac{f(u)}{u-z}\,du \quad \text{for all } 0<|z-z_0|<R,$$

so, define $$f(z_0)=\frac{1}{2\pi i}\cdot\int_C\frac{f(u)}{u-z_0}\,du,$$ we obtain the result.

Thm. Suppose that $f$ has an isolated singularity at $z_0$. TFAE:

1) For some $m \in \mathbb{N}$, $\lim_{z \to z_0} (z-z_0)^m \cdot f(z) = A \in \mathbb{C} \setminus \{0\}$. (That is $z_0$ is pole of order $m$).

2). $f(z) \to \infty$ as $z \to z_0$

3). $f(z) = g(z)/(z-z_0)^m$ for some analytic $g(z)$ and $m \in \mathbb{N}$, with $g(z_0) \neq 0$.

Proof. Suppose that there exists $m \in \mathbb{N}$ s.t. $\lim_{z \to z_0} (z-z_0)^m f(z)$ exists with non-zero.
Given $\varepsilon = |A|/2$, there exists $\delta > 0$ such that

$$|z-z_0| < \delta \Rightarrow |(z-z_0)^m \cdot f(z) - A| < |A|$$

So, using the triangle inequality for $0 < |z-z_0| < \delta$,

$$|A| - |z-z_0|^m \cdot |f(z)| < \frac{|A|}{2}$$

$$\Rightarrow \quad \frac{|A|}{2\delta^m} < \frac{|A|}{2|z-z_0|^m} < |f(z)|$$

Thus, taking $\delta \to 0$, then $f(z) \to \infty$. ] 1) $\Rightarrow$ 2).

Suppose that $f(z) \to \infty$ as $z \to z_0$. Put $g(z) = 1/f(z)$ in $B_\delta(z_0) \setminus \{z_0\}$ for some $\delta > 0$, which satisfies $f(z) \neq 0$. (If no such neibor exists, then $f$ is identically zero.)
Now, since $f(z) \to \infty$ as $z \to z_0$, so $g(z) \to 0$ as $z \to z_0$, so $z_0$ is removable to $g$.
Then, the Taylor expansion

$$g(z) = a_0 + a_1(z-z_0)^1 + \cdots \quad (|z-z_0| < \delta).$$

is valid. Since $g(z) \neq 0$ in $B_\delta(z_0) \setminus \{z_0\}$, so not all the coefficients are zero.
Let $m$ be the smallest integer s.t. $a_m \neq 0$. ($m \geq 1$ because $\lim_{z \to z_0} g(z) = a_0 = 0$).
So that $g(z) = a_m(z-z_0)^m + a_{m+1}(z-z_0)^{m+1} + \cdots$

Now, $$\frac{1}{(z-z_0)^m f(z)} = a_m + a_{m+1}(z-z_0) + \cdots \to a_m \text{ as } z \to z_0,$$

that is, $(z-z_0)^m f(z) \to a_m^{-1}$ as $z \to z_0$. ] 2) $\Rightarrow$ 3).

3) $\Rightarrow$ 2) is directly, the analytic $g$ is continuous at $z_0$.

「Isolated Singularity at $\infty$.

If $f$ is analytic for all $R < |z| < \infty$, for some $R > 0$, then we say that $f$ has an isolated singularity at $\infty$.

For this $R > 0$, observe that

$f(z)$ is analytic at $|z| > R$ if and only if $f\left(\frac{1}{z}\right)$ is analytic at $|z| < \frac{1}{R}$

$f(z)$ has an isolated singularity at $\infty$ iff $f\left(\frac{1}{z}\right)$ has an isolated singularity at $0$.

「Thm. Let $f$ be a non-constant entire function. Then,

$f$ is polynomial <u>if and only if</u> $f(z) \to \infty$ as $z \to \infty$.

Proof. Suppose that $f$ is a polynomial. WLOG, let $f(z) = a_0 + a_1 z + \cdots + a_n z^n$. $(a_n \neq 0)$.

Then, $f\left(\frac{1}{z}\right) = \frac{a_n}{z^n} + \frac{a_{n-1}}{z^{n-1}} + \cdots + \frac{a_1}{z} + a_0$ so that $\lim_{z \to 0} z^n \cdot f\left(\frac{1}{z}\right) = a_n \neq 0$,

So, $f(z)$ has a pole at $\infty$, so $\lim_{z \to \infty} f(z) = \infty$.

Conversely, suppose that: $f\left(\frac{1}{z}\right)$ has a pole of order $m$ at $0$. That is,

$$\lim_{z \to \infty} \frac{f(z)}{z^m} = A \in \mathbb{C} \setminus \{0\}.$$

That is, for some $R > 0$, $|z| \geq R \Rightarrow \left| \frac{f(z)}{z^m} - A \right| < 1$

$$\Rightarrow \left| \frac{f(z)}{z^m} \right| < 1 + |A|$$

$$\Rightarrow |f(z)| < (1 + |A|) \cdot z^m$$

So, $f$ is a polynomial of degree at most $m$.

Γ Singularities of rational functions.

Let $P$ and $q$ be polynomials over $\mathbb{C}$, $P(z)=a_0+a_1z+\cdots+a_nz^n$, $q(z)=b_0+\cdots+b_mz^m$.

and $f(z)=P(z)/q(z)$. Then, $(a_n, b_m \neq 0)$.

$f(z)$ has a removable singularity at $\infty$

iff $f(\frac{1}{z})$ has a removable singularity at $0$

iff for some $\varepsilon>0$ and $M>0$, $0<|\frac{1}{z}|<\varepsilon \Rightarrow |f(\frac{1}{z})|\leq M$.

iff 〃 $|z|>\frac{1}{\varepsilon} \Rightarrow |f(z)|\leq M$.

iff $\deg P(z) \leq \deg q(z)$.

$f(z)$ has a pole of order $k$ at $\infty$

iff $f(\frac{1}{z})$ has a pole of order $k$ at $0$.

iff $f\left(\frac{1}{z}\right)=\dfrac{P(\frac{1}{z})}{q(\frac{1}{z})}=\sum_{i=-k}^{\infty} c_i \cdot z^i$ for some $\{c_i\}_{i=-k}^{\infty}$

iff $P(\frac{1}{z})=\sum_{i=0}^{n} a_i \cdot \dfrac{1}{z^i}=\left(\sum_{i=0}^{m} b_i \cdot \dfrac{1}{z^i}\right)\cdot\left(\sum_{i=-k}^{\infty} c_i z^i\right)$

iff $n=m+k$, that is, the order $k=\deg P-\deg q>0$.

「Long division.

Consider a function represented by

$$f(z) = \left(\sum_{n=0}^{\infty} a_n z^n\right) \Big/ \left(\sum_{n=0}^{\infty} b_n z^n\right) = \frac{a_0 + a_1 z + a_2 z^2 + \cdots}{b_0 + b_1 z + b_2 z^2 + \cdots} \qquad (b_0 \neq 0).$$

Since $b_0 \neq 0$, this $f$ is analytic at $z=0$, so $f$ can be written as

$$f(z) = \frac{a_0 + a_1 z + \cdots}{b_0 + b_1 z + \cdots} = c_0 + c_1 z + \cdots$$

So that

$$\left(\sum_{n=0}^{\infty} a_n z^n\right) = \left(\sum_{n=0}^{\infty} b_n z^n\right) \cdot \left(\sum_{n=0}^{\infty} c_n z^n\right),$$ in some neighbor of the origin, thus

the uniqueness theorem gives

$$a_n = \sum_{k=0}^{n} b_k \cdot c_{n-k},$$ the Cauchy product.

「Example.

Let $f(z) = \dfrac{\pi \cdot \cot \pi z}{z^4}$. Then,

$$f(z) = \frac{\pi}{z^4} \cdot \left( \frac{1 - \frac{\pi^2}{2!} \cdot z^2 + \frac{\pi^4}{4!} \cdot z^4 + \cdots}{\pi z - \frac{\pi^3}{3!} \cdot z^3 - \frac{\pi^5}{5!} \cdot z^5 + \cdots} \right)$$

$$= \frac{1}{z^5} \cdot \left( \frac{1 - \frac{\pi^2}{2} z^2 + \frac{\pi^4}{4!} z^4 - \cdots}{1 - \frac{\pi^2}{3!} \cdot z^2 + \frac{\pi^4}{5!} \cdot z^4 - \cdots} \right)$$

$1 = 1 \cdot c_0 \Rightarrow c_0 = 1$

$a_1 = b_0 c_1 + b_1 c_0 = c_1 + 0 = 0. \Rightarrow c_1 = 0.$

$a_2 = b_0 c_2 + b_1 c_1 + b_2 c_0 = c_2 + 0 + -\frac{\pi^2}{6} \cdot 1 = -\frac{\pi^2}{2} \Rightarrow c_2 = -\frac{\pi^2}{3}.$

$a_3 = b_0 c_3 + b_1 c_2 + b_2 c_1 + b_3 c_0 = c_3 + 0 + -\frac{\pi^2}{6} \cdot 0 + 0 = 0 \Rightarrow c_3 = 0.$

$a_4 = b_0 c_4 + \cancel{b_1 c_3} + b_2 c_2 + \cancel{b_3} c_1 + b_4 c_0 = c_4 + c_2 \cdot -\frac{\pi^2}{6} + \frac{\pi^4}{5!} = c_4 + \pi^4 \cdot (1/18 + 1/120) = \frac{\pi^4}{24}.$

$\Rightarrow c_4 = \pi^4 \left( \frac{1}{24} - \frac{1}{18} - \frac{1}{120} \right) = \pi^4 \cdot \left( -\frac{1}{45} \right) = -\frac{1}{45} \pi^4.$

Consequently, the principal part of $f$ is

$$\frac{1}{z^5} - \frac{\pi^2}{3} \cdot \frac{1}{z^3} - \frac{\pi^4}{45} \cdot \frac{1}{z}.$$